Anthony Ng (linkedin.com/in/anthonytng | anthony.thomas.ng@gmail.com)

Education

M.S. in Robotics | Georgia Institute of Technology | Grad: Spring 2028
- Home school: Mechanical Engineering; MS-thesis
- Courses: Dynamics of Mechanical Systems, Linear Controls, Probability Theory for Sci/Eng

B.S. in Mechanical Engineering | University of Illinois Urbana-Champaign | Grad: Fall 2025
- GPA: 3.97/4.00 (Bronze Tablet, Highest Honors, James Scholar, 6x Dean's List); Minor: Computer Science
- Highlighted Coursework: DFM, Robot Dynamics and Controls, Computer Control of Mechanical Systems, Mechanical Design Optimization, Numerical Methods, Deep Learning for Computer Vision

Skills
- **Software:** Solidworks, Creo, Fusion 360, Arduino, Java, C++, Python, Matlab, MS Suite, Agile, Windchill
- **Other Skills:** DOE, 3D Printing & Machine Tools, Change Order/Doc Control, CT Scanning

Experience

Graduate Researcher | GT EPIC Lab (Dr. Aaron Young): Powered Prosthesis | Fall 2026 - Present
- Supporting prosthetic trials with motion-capture and onboard sensor data collection
- Integrating and configuring new BLDC actuators on Jetson Orin and E-stop for OSL powered prosthetic

Mechanical Engineering Intern | Intuitive Surgical | Summer 2026
- Designed and fabricated a modular fixture and test method to conduct component-level friction testing across all DaVinci Multiport Core 8 mm instruments w/ Solidworks
- Conducted CT scanning and material testing for metallic 3D printed vs metallic injection molded (MIM) patient-contacting components
- Designed and assembled safety cladding and fixturing for critical Force Feedback test equipment

Mechanical Engineering Intern | Brooks Automation | Summer 2025
- Led customer collaboration to diagnose EFEM vibrations using 3-axis accelerometer data and Python spectral analysis, presenting findings/recommendations to technical and non-technical stakeholders
- Performed static and dynamic load calculations and conducted FEA (including creating simplified CAD models) for various configurations in an upcoming robot redesign using Creo and Python
- Designed fixture to parallelize 24-hr motor test, allowing multiple units to be tested w/o full assembly

Mechatronic Equipment Engineering Intern | Intuitive Surgical | Summer 2024
- Designed, prototyped, and built fixture to improve safety and ergonomics in warehouse w/ Solidworks
- Collaborated with cross-functional team to design Da Vinci 5 surgeon-side console motor test module
- Finalized and refined designs leveraging FEA, 3D printing, and drawings (including GD&T)

Manufacturing Engineering Intern | Endovascular Engineering | Summer 2023

R&D Engineering Intern | Earlens Corporation | Summer 2022

Project Highlights

Oscillatory Flume Control Integration | UIUC ME470 (Senior Design) | Fall 2025
- Implemented a single-user-operable oscillatory flume in a team of 4 (demo video link)
- Assisted tuning for ODrive P + PI cascade position controller to achieve square, triangle, and sine waves with amplitude range 10-30 cm and frequency range 0.1-0.5 Hz
- Designed (Fusion360) piston drive & extrusion frame, using only 3D-printed, waterjet, and COTS parts
- Awarded 2nd out of 32 teams for Best Trade Show Exhibit (faculty-voted)

3-Link Manipulator Obstacle Course | UIUC ME446 (Robot Dynamics and Controls) | Spring 2025
- Implemented task-space PD & impedance control with variable stiffness for compliant interaction and obstacle navigation on a CRS robot arm using C and MatLab in a team of 2 (demo video link)
- Developed forward/inverse kinematics with Jacobian force mapping for real-time 3D trajectory tracking
- Programmed straight-line trajectory generation to traverse through peg insertion, zig-zag corridors, and pressing an egg within an applied load range of 500-1000 g